part of the design phase commissioning provider team. (See the Selection of the CxP Team section later in this chapter.) A typical design review process is as follows:

- Written comments from the reviewers are provided to the design team and owner.
- Comments are reviewed, and the design team returns written responses.
- Meetings are scheduled with the review team and the designers to resolve comments, allowing the owner to understand the issues and have an opportunity to provide direction as needed.
- Design concerns, comments, and actions taken are recorded in the design review document. Changes are made as agreed, and the commissioning review team verifies the change and closes the issue as appropriate.

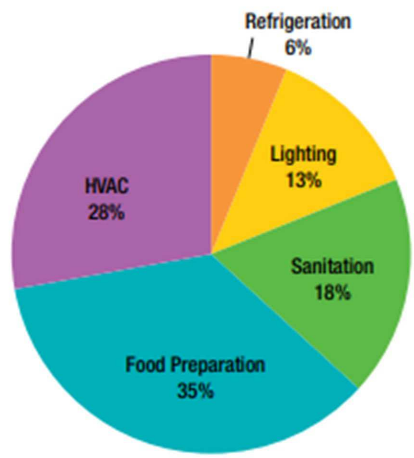
Using a best-practices approach, the commissioning design review could occur four different times during the design: at 100% of schematic design, at 100% of design development, at 50% of complete construction documents, and at approximately 95% of construction documents. (Combining the review of the first two phases can be done on smaller projects.) An advantage of four reviews, however, is that the design is evaluated based on the OPR document goals before the next phase starts.

Although four reviews are ideal, particularly for complicated projects, having at least two (as required for commissioning by Standard 189.1) provides at least some level of checks and balances.

Changes to optimize building performance, daylighting considerations, system selection, and stacking/massing synergies can best be addressed during schematic design review. Review during design development allows the team to identify potential problems and constructability perspectives early enough to resolve many issues before the construction documents phase starts.

The quantity of design concerns typically increases during the 95% construction document phase because more detail is provided about each building system and component. The concerns identified at that point typically revolve around details—finishes, coordination conflicts, and so on. Resolving these concerns provides clearer direction to the contractors, resulting in better cost and schedule predictions.

It is not uncommon for concerns identified at the 50% construction documents phase to go unaddressed by a design team. This is especially true in fast-track projects when designers, responding to owner and contractor demands for documents, struggle to finish and deliver their work product. This is why the later 95% review is so important.



Source: U.S. EPA. (2010). *ENERGY STAR® Guide for Restaurants - Putting Energy into Profit.* (Figure reproduced)

Figure 9-1 Breakdown of energy cost by kitchen functional area.

Energy Star commercial kitchen appliances can save between 10% to 65% over conventional appliances depending on the type.

- **Ventilation and HVAC:** Consider the use of demand based kitchen exhaust systems that are exhaust systems that vary the exhaust airflow from the hoods based on the intensity of the cooking based on smoke and heat sensors. Also, consider the use of side panels on hoods to cut down spillage without the need to increase exhaust airflow.
- **Refrigeration:** Consider the use of Energy Star qualified commercial refrigerators and freezers which are more energy-efficient because they have components such as electronically commutated motors (ECMs), evaporator and condenser fan motors, hot gas antisweat heaters or high-efficiency compressors. Energy Star qualified commercial refrigerators and freezers can reduce energy costs by as much as 35%.
- **Sanitation:** Consider the use of Energy Star qualified dishwashers which use less energy and less water. When a dishwasher uses less water, energy is also saved by heating less water at the building and booster heater levels More than 75% of dishwasher operating costs are for heating water with central and booster heaters. Energy Star qualified commercial dishwashers use approximately 25% less energy and water than standard models.
- **Lighting:** The typical commercial kitchen has lights on for 16 to 20 hours per day so consider the use of high efficiency LED lights.

ASHRAE GreenTip #9-15

Automated Refrigerant Leak Detection and Recovery

GENERAL DESCRIPTION

In the past, leaking chillers were acceptable because of the ready supply and the low cost of refrigerant. Today if a refrigerant leaks in chillers (be it R134a, R410a, HFOs), it not only cost the building owner but also the refrigerant released to the atmosphere will cause global warming, increase carbon footprint and in a way affect climate change. This may not be acceptable under the US Clean Air Act.

Automated refrigerant leakage detection and recovery system not only prevent further leakage of refrigerant being dispersed into the atmosphere but also prevent wastage of refrigerant, achieve cost savings, minimize risk and protect the safety of mechanical room personnel and the environment. If owners choose to follow any green building certification, automated refrigerant leakage detection can benefit them not just by being environmentally sound but also by lowering operation cost, reducing carbon footprint, reducing liability and risk, enhancing marketability through higher return on investment and increasing tenant rental.

WHEN/WHERE IT'S APPLICABLE

Automated refrigerant leakage detection and recovery systems are typically used in mechanical rooms where chillers are installed. Some chillers lose 2% to 15% of their refrigerant charge annually through different mechanisms. In the past, all of the lost refrigerant was released to the atmosphere.

Refrigerant loss occurs primarily through four different channels: leaks, poor service procedures, contamination, and purge cycles. Depending on the local service standards, the relative magnitude of these channels can vary widely.

The case of high-pressure chillers is slightly different than low pressure chillers. High pressure chillers maintain positive pressure throughout the refrigerant circuit; therefore, they do not require purge units. Aside from this difference, the refrigerant loss mechanisms are basically the same as in low pressure chillers. All of the losses for both high- and low-pressure chillers can be significantly tionalized into the building maintenance culture. For example, consider the case of a chiller staging sequence using supply and return temperatures and flowmeter(s). These sensors may drift over time, and typical accepted error ranges in these types of sensors will yield inaccurate load calculations that may disrupt proper staging. This strategy can result in high overall efficiency, but to achieve this high level of efficiency, this requires a regular calibration and maintenance checks to maintain this high efficiency.

Requirements for System Architecture Rationale. It is important to ensure that, in the requirements for the controls submittal, the controls contractor is mandated to provide calculations and rationale for the number and layout of the primary (peer-to-peer) and local (application-specific) controllers in relation to the total number of points and other network traffic. It should also be required that the contractor describe how many points can be reasonably trended simultaneously without appreciably affecting point value refresh rates, and describe the impacts on network speed that alternative layouts would have (or alternatively provide insight into the network requirements needed to operate the desired system architecture).

The building automation system (BAS) performance requirements should be defined in the contract documents (i.e., specifications such as those set forth in ASHRAE Guideline 13-2015, *Specifying Direct Digital Control Systems* [ASHRAE 2014]). The performance requirements are defined during the predesign phase and should be contained in the OPR document in accordance with ANSI/ASHRAE/IES Standard 202-2018.

Requirements for Clear Control Sequences. Ensure that the requirements for the controls submittals in the specifications include statements requiring the following:

- A brief overview narrative of the system, generally describing its purpose, components, and function (i.e., the design intent for the controls)
- All interactions with other systems
- Detailed delineation of interaction between any localized controls and the BAS, listing what points only the BAS monitors and what BAS points are control points and are thus adjustable
- Start-up, warm-up, cooldown, occupied, and unoccupied operating modes, plus power failure recovery and alarm sequences
- Capacity control sequences and equipment staging
- Initial and recommended values for all adjustable settings, set points, and parameters that are typically set or adjusted by building operators, as well as any other control settings, fixed values, delays, and so on, that will be useful during testing and ongoing operating the equipment
- Rough zone analysis requirements to assure reset strategies are effective
- Energy mapping requirements to help operators understand building efficiency
- System override capabilities and requirements

strategy, control capabilities, annual service requirements, and upfront costs, including ductwork installation.

Selecting an HRV or ERV depends on the climate and the specific needs or conditions of the dwelling. A general rule to keep in mind is that energy always transfers from greater to lesser. That is, higher temperature transfers to lower temperature and likewise for moisture differences. Generally, an ERV is better for hot and humid locales because it directs some of the humidity back to the outdoors rather than bringing it all indoors, as an HRV will do. Using the same rationale, if a dwelling is too humid indoors during the winter heating season, an HRV is a better choice for ridding the dwelling of this moisture. Generally, ERVs are a better choice for hot and humid climates, and HRVs are better for cold and dry climates, but other factors are often more important than the climate, such as dwelling tightness level and the number of occupants.

VENTILATION RATE

All indoor spaces benefit from increased ventilation based on the assumption that the outdoor air is healthier than the indoor air; many studies have demonstrated that this is the case. The biggest unanswered question is how much ventilation does the space need or, better stated, what ventilation rate will achieve acceptable indoor air quality. This ventilation rate is typically determined by code, either the *International Residential Code* (IRC) or ASHRAE Standard 62.2, *Ventilation and Indoor Air Quality in Residential Buildings*. If there is no enforceable ventilation code in a location, it is recommended that ASHRAE Standard 62.2 be followed.

SOURCES OF FURTHER INFORMATION

ASHRAE. 2022. ASHRAE Standard 62.2-2022, *Ventilation and Indoor Air Quality in Residential Buildings*. Peachtree Corners, GA: ASHRAE.

US Environmental Protection Agency https://www.epa.gov/indoor-air-quality-iaq/improving-indoor-air-quality.